SHRI RAMSWAROOP MEMORIAL UNIVERSITY

PROBLEM SET – I

Session	: 2025-26(Even Sem.)	**Semester**	: VI
Course	: BCA & B.Sc IT	**Subject Code**	:UCS6124
Branch	: Normal & Specialization	**Subject Name**	: Machine Learning Elective-II
Group	: ALL	**Name of Faculty Member**	: Dr. Archana Sachindeo Maurya, Er. Aakriti Nigam, Er. Prem Kumar Yadav
Unit	: I	**Topic Covered**	: Machine Learning Foundations

S. No.	Questions
1	Explain the need for mathematics in Machine Learning and justify why it is used in ML algorithms.
2	How is Machine Learning different from traditional programming approaches? Explain.
3	Describe the various stages involved in the Machine Learning pipeline in detail with a neat diagram.
4	Explain how Linear Algebra forms the mathematical foundation of Machine Learning operations, with emphasis on matrix computations and vector spaces.
5	Differentiate between Probability Theory and Statistics with examples.
6	Explain the need for mathematics in Machine Learning and justify why it is used in ML algorithms.
7	Why do we need Machine Learning? Explain the importance of data in Machine Learning systems.
8	Explain curve fitting and its significance in Machine Learning. Discuss underfitting and overfitting
9	Discuss the popular tools and software frameworks used in Machine Learning along with their applications.
10	Analyze the role of Decision Theory and Information Theory in building Machine Learning models.

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PROBLEM SET – II

Session	: 2025-26(Even Sem.)	**Semester**	: VI
Course	: BCA & B.Sc IT	**Subject Code**	:UCS6124
Branch	: Normal & Specialization	**Subject Name**	: Machine Learning Elective-II
Group	: ALL	**Name of Faculty Member**	: Dr. Archana Sachindeo Maurya, Er. Aakriti Nigam, Er. Prem Kumar Yadav
Unit	: II	**Topic Covered**	: Supervised Learning-I

<table>
<tr><th>S. No.</th><th>Questions</th></tr>
<tr><td>1</td><td>Derive the Linear Regression model using the Least Squares estimation method and critically analyze the conditions under which overfitting arises in linear regression models.</td></tr>
<tr><td>2</td><td>Explain Linear Basis Function Models and examine polynomial basis functions with a suitable example, highlighting their impact on model complexity and generalization.</td></tr>
<tr><td>3</td><td>Describe Bayesian Linear Regression and critically compare it with Ordinary Least Squares (OLS) regression in terms of uncertainty modeling and parameter estimation.</td></tr>
<tr><td>4</td><td>Define a discriminant function in classification and analyze its role in supervised learning algorithms with suitable examples.</td></tr>
<tr><td>5</td><td>Compare and contrast generative and discriminative models in machine learning, emphasizing their learning objectives, assumptions, and applications. Provide two examples of each.</td></tr>
<tr><td>6</td><td>Illustrate Logistic Regression in detail and critically analyze how it differs from Linear Regression with respect to model formulation and output interpretation.</td></tr>
<tr><td>7</td><td>Derive the Sigmoid function used in Logistic Regression and analyze its significance in probabilistic classification and decision boundary formation.</td></tr>
<tr><td>8</td><td>Explain Logistic Regression in detail and critically analyze how it differs from Linear Regression with respect to model formulation and output interpretation.</td></tr>
<tr><td>9</td><td>Given the dataset:
<table>
<tr><td>Feature</td><td>Class A</td><td>Class B</td></tr>
<tr><td>P(x1)</td><td>0.6</td><td>0.4</td></tr>
<tr><td>P(x2)</td><td>0.7</td><td>0.3</td></tr>
</table>
Class Prior Probabilities:
P(A)=0.5
P(B)=0.5
1. Compute P(A | x1, x2)
2. Compute P(B | x1, x2)
3. Classify the sample using Naïve Bayes rule.</td></tr>
<tr><td>10</td><td>Calculate Maximum Likelihood Estimation (MLE) for the following observations:
X=[2, 4, 6, 8]
1. Estimate the mean using MLE
2. Estimate the variance using MLE
3. State the assumptions made.</td></tr>
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PROBLEM SET – III

Session	: 2025-26(Even Sem.)	**Semester**	: VI
Course	: BCA & B.Sc IT	**Subject Code**	:UCS6124
Branch	: Normal & Specialization	**Subject Name**	: Machine Learning Elective-II
Group	: ALL	**Name of Faculty Member**	: Dr. Archana Sachindeo Maurya, Er. Aakriti Nigam, Er. Prem Kumar Yadav
Unit	: III	**Topic Covered**	: Supervised Learning-II

<table>
<tr><th>S. No.</th><th>Questions</th></tr>
<tr><td>1</td><td>Analyze how margin maximization affects the generalization bound using VC-dimension or structural risk minimization.</td></tr>
<tr><td>2</td><td>Formulate the soft-margin SVM optimization problem and explain the role of slack variables.</td></tr>
<tr><td>3</td><td>Analyze the impact of kernel choice on decision boundaries and computational complexity. Also discuss limitations of kernel methods for large-scale datasets.</td></tr>
<tr><td>4</td><td>Compare the computational and numerical advantages of dual vs primal optimization for sparse and dense datasets.</td></tr>
<tr><td>5</td><td>Compare RBF networks with multilayer perceptron’s (MLPs) in terms of learning dynamics, locality, and convergence.</td></tr>
<tr><td>6</td><td>Analyze k-NN from a non-parametric learning perspective. Discuss the curse of dimensionality and propose advanced strategies to mitigate it.</td></tr>
<tr><td>7</td><td>Analyze how depth, width, and activation functions influence the expressive power of feed-forward networks.</td></tr>
<tr><td>8</td><td>Derive the backpropagation equations using the chain rule for a multi-layer network.</td></tr>
<tr><td>9</td><td>What is regularization in neural networks? Explain its role in preventing overfitting.</td></tr>
<tr><td>10</td><td>Numerical on Instance-based Learning(K-NN)
Given the dataset:
<table>
<tr><td>Point</td><td>X1</td><td>X2</td><td>Class</td></tr>
<tr><td>P1</td><td>1</td><td>2</td><td>A</td></tr>
<tr><td>P2</td><td>2</td><td>3</td><td>A</td></tr>
<tr><td>P3</td><td>3</td><td>3</td><td>B</td></tr>
<tr><td>P4</td><td>6</td><td>5</td><td>B</td></tr>
</table>
For query point Q=(3,2):
1. Compute Euclidean distance.
2. Predict the class for:
• K=1
• K=3
3. Analyze how distance weighting changes the prediction.</td></tr>
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